

Synthesis of New Pyridyl Alcohols and Their Use as Catalysts in the Asymmetric Addition of Diethylzinc to Aldehydes

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ABSTRACT Optically active new pyridyl alcohols **1–4**, which can be easily synthesized by the reaction of (+)-camphor or (–)-menthone with lithiated pyridine derivatives, were applied as chiral ligands in the asymmetric addition of diethylzinc to aldehydes. Good yields with up to 94.0% enantiomeric excesses were observed in these reactions. *Chirality* 14:28–31, 2002. © 2002 Wiley-Liss, Inc.

KEY WORDS: asymmetric addition; pyridyl alcohols; (+)-camphor; (–)-menthone; diethylzinc

Catalytic asymmetric synthesis is a valuable method for the preparation of optically active substances. Among them, the stereoselective addition of organometallics to one of the two enantiotopic faces of a carbonyl group has been extensively studied, especially the enantioselective addition of dialkylzinc to aldehydes in the presence of chiral catalyst.^{1–3} It is well known that organozinc compounds are inert to the carbonyl group. The discovery that certain β -aminoalcohols can promote the ethylation of benzaldehyde with diethylzinc initiated a search for efficient chiral auxiliaries since Oguni's first report in 1984.⁴ Many chiral ligands (most are aminoalcohols, particularly β -aminoalcohols) have been synthesized and tested.^{1–6} Only a few examples demonstrated the syntheses and efficient uses of γ -aminoalcohols^{7,8} and δ -aminoalcohols.^{9–13} Pyridyl alcohols, which have a structure similar to β -aminoalcohols, have also been applied efficiently in these reactions.^{14,15} However, few chiral pyridyl alcohols with a structure similar to γ -aminoalcohols have been synthesized and used to catalyze the asymmetric addition of diethylzinc to aldehydes.^{16–18} In this report, new pyridyl alcohols with this structure and their application to promote the enantioselective ethylation of aldehydes is described.

EXPERIMENTAL SECTION

All reactions were carried out under Ar atmosphere. Melting points were measured on a Kofler melting apparatus and are uncorrected. ¹H and ¹³C NMR spectra were measured with a Bruker AM-400 NMR spectrometer with TMS as an internal reference. Infrared spectra were recorded on a Nicolet 170SX FT-IR spectrometer. Electron ionization mass spectra were obtained on a Hewlett-Packard HP5988A mass spectrometer. Positive ion FAB mass

spectra as 3-nitrobenzyl alcohol matrix were recorded on a VG ZAB-HS mass spectrometer. Elemental analyses were performed on a Carlo-Erba-1106 elemental analyzer. Optical rotations were measured on a JASCO J-20C automatic polarimeter. Enantiomeric excess (e.e.) determination was carried out on Varian CP-3380 GC instrument with FID as the detector, nitrogen as carrier gas, and a Chrompack CP-Chirasil-DEX CB capillary column, or on Varian SD-200 HPLC instrument with UV detector, hexane/2-propanol as eluent, and a Chiralcel-OD column. All solvents used were dried by standard methods and aldehydes were purified using standard, published methods before use.

Preparation of Chiral pyridyl Alcohol **1**

To a 100 ml flame-dried three-neck flask was added 2, 6-lutidine (1.16 ml, 10.0 mmol) and anhydrous ether (30 ml). The solution was cooled to 0°C and 2.4 M butyllithium in hexane (4.3 ml, 10.3 mmol) was added using a pressure-equalizing dropping funnel over 15 min with stirring. The cooling bath was removed and the solution was allowed to stir for 1 h while the temperature rose to ambient. A solution of (+)-camphor (1.52 g, 10.0 mmol) in ether (20 ml) was added dropwise over 15 min with vigorous stirring while the temperature cooled to 0°C. The mixture was stirred for additional 2 h and hydrolyzed with saturated aqueous NH₄Cl solution. When decomposition was completed, the ether layers were separated and the water layer was extracted with ethyl acetate (3 × 20 ml). The combined organic layers were washed with brine, dried over anhy-

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Received for publication 24 April 2001; Accepted 18 June 2001

drous Na_2SO_4 , and the solvent was evaporated under reduced pressure. The crude product was purified by flash chromatography (ethyl acetate : petroleum 1 : 4) to give a white solid (2.41 g, 93% yield): Mp 28–29°C; $[\alpha]_D^{20} = -19.4$ (c 1.08, CHCl_3); IR (KBr, cm^{-1}): 3329–3347, 1572, 1550; ^1H NMR (CDCl_3 , δ ppm): 0.51 (s, 3H), 0.82 (s, 3H), 1.01–1.12 (m, 1H), 1.13 (s, 3H), 1.33–1.48 (m, 3H), 1.68–1.74 (m, 2H), 2.03–2.06 (m, 1H), 2.49 (s, 3H), 2.94 (s, 2H), 6.74 (br, 1H), 6.96–7.02 (m, 2H), 7.51 (m, 1H). ^{13}C NMR (CDCl_3 , δ ppm): 11.20, 20.99, 21.42, 24.19, 27.18, 30.78, 43.92, 44.99, 47.44, 49.55, 52.33, 81.14, 120.91, 121.21, 132.01, 156.65, 159.74; MS m/z (EI): 259 (M^+ , 3), 244 (3), 188 (3), 176 (6), 149 (33), 107 (100), 95 (88); positive ion FAB mass spectra, m/z : 260 ($\text{M}^+ + \text{H}$); Anal. calcd for $\text{C}_{17}\text{H}_{25}\text{NO}$: C 78.72, H 9.71, N 5.40; Found: C 78.63, H 9.65, N 5.49.

Preparation of Chiral Pyridyl Alcohol 2

Prepared from (–)-menthone (1.54 g, 10.0 mmol) and 2,6-lutidine (1.16 ml, 10.0 mmol) in a manner similar to that described above to give a white solid (2.41 g, 92% yield): Mp 37–38°C; $[\alpha]_D^{20} = -86.0$ (c 1.36, CHCl_3); IR (KBr, cm^{-1}): 3343–3355, 1577, 1544; ^1H NMR (CDCl_3 , δ ppm): 0.72 (d, $J = 6.5$ Hz, 3H), 0.92 (d, $J = 6.8$ Hz, 3H), 0.99 (d, $J = 6.8$ Hz, 3H), 1.05–1.09 (m, 1H), 1.16–1.21 (m, 1H), 1.32–1.40 (m, 1H), 1.47–1.64 (m, 3H), 1.70–1.78 (m, 2H), 2.20–2.26 (m, 1H), 2.47 (d, $J = 14.0$ Hz, 1H), 2.50 (s, 3H), 3.35 (d, $J = 14.0$ Hz, 1H), 5.60 (br, 1H), 6.89 (d, $J = 7.8$ Hz, 1H), 6.98 (d, $J = 7.8$ Hz, 1H), 7.48 (m, 1H); ^{13}C NMR (CDCl_3 , δ ppm): 18.09, 20.90, 22.42, 23.82, 26.25, 27.69, 35.41, 43.36, 46.15, 47.51, 51.20, 74.77, 120.63, 121.54, 136.63, 157.12, 159.95; MS m/z (EI): 261 (M^+ , 2), 246 (3), 204 (3), 176 (22), 107 (100); positive ion FAB mass spectra, m/z : 262 ($\text{M}^+ + \text{H}$); Anal. calcd for $\text{C}_{17}\text{H}_{27}\text{NO}$: C 78.10, H 10.42, N 5.36; Found: C 78.02, H 10.47, N 5.29.

Preparation of Chiral Pyridyl Alcohol 3

Prepared from (+)-camphor (1.52 g, 10.0 mmol) and 2-picoline (0.99 ml, 10.0 mmol) in a manner similar to that described for pyridyl alcohol 1 to give a white solid (2.25 g, 92% yield): Mp 43–44°C; $[\alpha]_D^{20} = -21.0$ (c 1.37, CHCl_3); IR (KBr, cm^{-1}): 3337–3352, 1570, 1542; ^1H NMR (CDCl_3 , δ ppm): 0.40 (s, 3H), 0.72 (s, 3H), 0.97–1.03 (m, 1H), 1.04 (s, 3H), 1.27–1.40 (m, 3H), 1.59–1.66 (m, 2H), 1.94–2.00 (m, 1H), 2.85–2.90 (AB system, $J = 14.2$ Hz, 2H), 6.25 (br, 1H), 7.04–7.10 (m, 2H), 7.52 (m, 1H), 8.34 (m, 1H); ^{13}C NMR (CDCl_3 , δ ppm): 11.05, 20.89, 21.28, 27.05, 30.62, 44.85, 45.05, 47.28, 49.42, 52.19, 81.04, 121.31, 124.24, 136.67, 147.79, 160.38; MS m/z (EI): 245 (M^+ , 3), 230 (3), 188 (1), 174 (2), 162 (4), 107 (39), 93 (100); positive ion FAB mass spectra, m/z : 246 ($\text{M}^+ + \text{H}$); Anal. calcd for $\text{C}_{16}\text{H}_{23}\text{NO}$: C 78.31, H 9.45, N 5.71; Found: C 78.41, H 9.40, N 5.62.

Preparation of Chiral Pyridyl Alcohol 4

Prepared from (–)-menthone (1.54 g, 10.0 mmol) and 2-picoline (0.99 ml, 10.0 mmol) in a manner similar to that described for pyridyl alcohol 1 to give a white solid (2.33 g, 94% yield): Mp 73–74°C; $[\alpha]_D^{20} = -62.9$ (c 1.03, CHCl_3); IR

(KBr, cm^{-1}): 3324–3366, 1566, 1520; ^1H NMR (CDCl_3 , δ ppm): 0.72 (d, $J = 6.3$ Hz, 3H), 0.95 (d, $J = 6.8$ Hz, 3H), 1.01 (d, $J = 6.8$ Hz, 3H), 1.05–1.13 (m, 1H), 1.15–1.21 (m, 1H), 1.26–1.41 (m, 1H), 1.48–1.65 (m, 3H), 1.72–1.77 (m, 2H), 2.26–2.29 (m, 1H), 2.53 (d, $J = 13.8$ Hz, 1H), 3.43 (d, $J = 13.8$ Hz, 1H), 4.91 (br, 1H), 7.14–7.27 (m, 2H), 7.63 (m, 1H), 8.50 (m, 1H); ^{13}C NMR (CDCl_3 , δ ppm): 18.11, 19.36, 22.42, 23.83, 26.31, 27.73, 35.38, 46.53, 47.42, 51.14, 74.88, 121.21, 124.85, 136.76, 148.22, 160.55; MS m/z (EI): 247 (M^+ , 1), 232 (2), 186 (4), 177 (2), 162 (19), 93 (100); positive ion FAB mass spectra, m/z : 248 ($\text{M}^+ + \text{H}$); Anal. calcd for $\text{C}_{16}\text{H}_{25}\text{NO}$: C 77.68, H 10.19, N 5.66; Found: C 77.76, H 10.11, N 5.59.

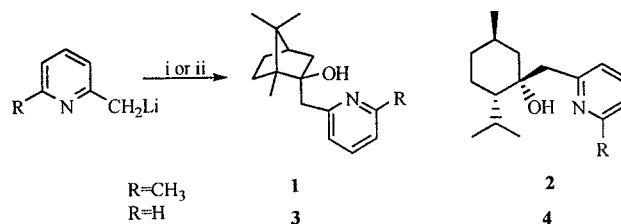
General Procedure for the Asymmetric Addition of Diethylzinc to Benzaldehyde

To a solution of catalyst 4 (1.0 mmol) in toluene (4 ml) and hexane (4 ml) at 0°C was added a 1.0 M solution (10.0 ml, 10.0 mmol) of diethylzinc in hexane. After stirring for 30 min at 0°C, freshly distilled benzaldehyde (5.0 mmol) was added. The mixture was stirred at 0°C until the reaction completed. After addition of 1 N HCl (10 ml), the phases were separated. The water layer was extracted with ethyl acetate (3 × 20 ml). The combined organic layers were washed with brine, dried over anhydrous Na_2SO_4 , and concentrated under reduced pressure. After purification by flash chromatography (ethyl acetate : petroleum 1 : 5), the e.e. was determined by GC.

RESULTS AND DISCUSSION

Pyridyl alcohols 1–4¹⁸ can be easily synthesized by the reaction of (+)-camphor and (–)-menthone with 2-lithiomethyl-6-methyl-pyridine or 2-picolylithium^{19,20} to give a single diastereomer as shown by ^1H NMR analyses of the products (Scheme 1). After the usual work-up and purification, they were obtained as white solid in high yields.

As a prototype reaction, the addition of diethylzinc to benzaldehyde at 0°C was used to test ligands 1–4; the results are summarized in Table 1. Initially, the reaction was carried out in toluene/hexane (1/1, v/v) with 20 mol% of ligands. They catalyze the addition of diethylzinc to benzaldehyde efficiently and afford 1-phenyl-propanol in high yields with low to good e.e. (38.1–92.3%) (entries 1 and 5–7). By varying the catalysts loading of ligands 1 and 4, the yields and e.e. values decreased with the decreasing of the amount of the chiral catalysts (entries 1–4, 7, 16–18). This is reasonable because aminoalcohols have



Scheme 1. Reagents and conditions: (+)-camphor, ether, 0°C; (–)-menthone, ether, 0°C.

TABLE 1. Enantioselective addition of diethylzinc to benzaldehyde promoted by pyridyl alcohols 1–4^a

Entry	Ligand (mol%)	Solvent	Yield (%) ^b	E.e. (%) (Config.) ^c
1	1 (20)	Toluene/Hexane (1/1, v/v)	91	74.8 (<i>R</i>)
2	(10)	Toluene/Hexane (1/1, v/v)	93	68.7 (<i>R</i>)
3	(5)	Toluene/Hexane (1/1, v/v)	90	63.1 (<i>R</i>)
4	(2.5)	Toluene/Hexane (1/1, v/v)	88	56.0 (<i>R</i>)
5	2 (20)	Toluene/Hexane (1/1, v/v)	86	68.8 (<i>S</i>)
6	3 (20)	Toluene/Hexane (1/1, v/v)	93	38.1 (<i>R</i>)
7	4 (20)	Toluene/Hexane (1/1, v/v)	94	92.3 (<i>S</i>)
8	(20)	Hexane	92	89.6 (<i>S</i>)
9	(20)	Benzene/Hexane (1/1, v/v)	93	90.1 (<i>S</i>)
10	(20)	Toluene	90	87.9 (<i>S</i>)
11	(20)	Benzene	87	83.6 (<i>S</i>)
12	(20)	THF	49	65.5 (<i>S</i>)
13	(20)	Dichloromethane	63	69.6 (<i>S</i>)
14	(20)	Acetonitrile	58	78.5 (<i>S</i>)
15	(20)	Ether	61	74.3 (<i>S</i>)
16	(10)	Toluene/Hexane (1/1, v/v)	91	90.4 (<i>S</i>)
17	(5)	Toluene/Hexane (1/1, v/v)	90	85.6 (<i>S</i>)
18	(2.5)	Toluene/Hexane (1/1, v/v)	85	76.0 (<i>S</i>)

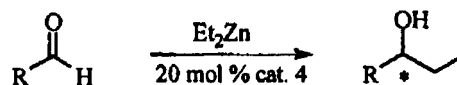
^aThe reactions were carried out in indicated solvent at 0°C. Benzaldehyde/Et₂Zn = 5.0/10.0 (mmol).^bBased on isolated product.^cThe e.e. values were determined by GC with Chrompack CP-Chirasil-DEX CB capillary column and the configuration was determined by comparison of the specific rotation with the known compound.

the so-called “ligand acceleration effect.”²¹ We also investigated the effect of solvents with ligand **4** as catalyst and found that they drastically affected the e.e. values and yields of the asymmetric addition reaction (entries 7–15). Hexane, benzene, toluene, benzene/hexane, and toluene/hexane are good solvents for achieving high yields and e.e. values (entries 7–11). Poor yields and e.e. values were observed using THF, dichloromethane, acetonitrile, or ether as solvents (entries 12–15).

The substituents at the hydroxy-bearing carbon, nitrogen-bearing carbon, or nitrogen atom of chiral aminoalcohol play an important role in the asymmetric induction. It is known that when the substituents at the hydroxy-bearing carbon of aminoalcohol become more bulky, the degree of asymmetric induction becomes higher.²² In spite of the presence of the *tert* alcohol moiety in ligands **1–4**, great differences of enantioselectivity have been observed under the optimized reaction conditions because of the difference of substituents on the hydroxy-bearing carbon or pyridine ring and ligand **4** gave the best result.

It can be seen from the results that the enantioselectivity of the reaction is very sensitive to the structure of the chiral catalyst and the configurations of the products are controlled essentially by the chirality of the hydroxy-bearing carbon of the pyridyl alcohols. For pyridyl alcohols **1** and **3** with an (*S*)-configuration of the hydroxy-bearing carbon, which were derived from (+)-camphor, (*R*)-1-phenyl-propanol was produced (entries 1–4, 6). While for pyridyl alcohols **2** and **4** with an (*R*)-configuration the hydroxy-bearing carbon, which was derived from (–)-menthone, (*S*)-1-phenyl-propanol was obtained (entries 5, 7–18). The highest e.e. (92.3%) was realized by carrying out the reaction in toluene/hexane (1/1, v/v) at 0°C using 20 mol% of pyridyl alcohol **4** (entry 7).

The enantioselective addition of diethylzinc to both aromatic and aliphatic aldehydes catalyzed by chiral ligand **4** was further examined and the results are summarized in Table 2. In all cases, moderate to good enantioselectivities

TABLE 2. Enantioselective addition of diethylzinc to aldehydes promoted by pyridyl alcohol **4^a**

Entry	Substrate	Yield (%) ^b	E.e. (%) (Config.) ^c
1	Benzaldehyde	94	92.3 (<i>S</i>)
2	<i>o</i> -Anisaldehyde	92	84.1 (<i>S</i>)
3	<i>p</i> -Anisaldehyde	90	89.7 (<i>S</i>)
4	<i>p</i> -Tolualdehyde	91	88.6 (<i>S</i>)
5	<i>o</i> -Chlorobenzaldehyde	82	83.2 (<i>S</i>)
6	<i>p</i> -Chlorobenzaldehyde	87	88.4 (<i>S</i>)
7	3,4-Dimethoxybenzaldehyde	97	88.7 (<i>S</i>) ^d
8	4-(Dimethylamino)benzaldehyde	98	76.5 (<i>S</i>) ^d
10	1-Naphthaldehyde	92	94.0 (<i>S</i>) ^d
11	2-Naphthaldehyde	90	82.7 (<i>S</i>) ^d
12	<i>trans</i> -Cinnamaldehyde	87	76.3 (<i>S</i>) ^d
13	Dodecylaldehyde	83	59.6 (<i>S</i>) ^e
14	1-Nonanal	86	54.2 (<i>S</i>) ^e
15	Cyclohexanecarboxaldehyde	80	61.3 (<i>S</i>) ^e

^aThe reactions were carried out in toluene/hexane (1/1, v/v) at 0°C with catalyst/aldehyde/Et₂Zn = 1.0/5.0/10.0 (mmol).^bBased on isolated product.^cExcept as noted, the e.e. values were determined by GC with Chrompack CP-Chirasil-DEX CB capillary column and the configurations were determined by comparison of the specific rotation with the known compounds.^dThe e.e. values were determined by HPLC with a Chiralcel-OD column.^eThe e.e. values were determined by GC after acetylation.

(54.2–94.0%) were obtained in high yields. It seems that the enantioselectivity for *para*-substituted aromatic aldehydes is higher than that for *ortho*-substituted one and there is no significant electronic influence of substrate on the enantioselectivity. Aromatic aldehydes were ethylated enantioselectively to produce the corresponding alcohol with relatively high e.e. values (entries 1–11) and the best result was obtained by the reaction of 1-naphthaldehyde to yield the corresponding alcohol with 94.0% e.e. (entry 10). However, for aliphatic aldehydes the enantioselectivities were moderate (entries 12–15).

In conclusion, pyridyl alcohol **4**, which was easily synthesized in one step from (–)-menthone and 2-picolyllithium, is an efficient chiral ligand for the asymmetric addition of diethylzinc to aldehydes. Further studies directed toward the application of these ligands in other enantioselective reactions and the syntheses of more efficient ligands are in progress.

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